ETC3250/5250 Introduction to Machine Learning

Jack Jewson

Outline

Welcome! Meet the teaching team

Chief examiner: Jack Jewson

Communication: All questions need to be communicated through the Moodle Discussion forum. **I will not respond to emails about module content.** Any of a private matter can be addressed to etc3250-etc5250@monash.edu.

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- Harriet: 4th year PhD student working on visualisation of uncertainty
- Jayani: 4th year PhD student working on methods for understanding how non-linear dimension reduction warps your data

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- Wednesdays 10am-11am - Consultation: Chance to ask me for **clarifications**

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- manage very large data sets in a modern software environment.
- explain and interpret the analyses undertaken clearly and effectively.

Assessment

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Weekly learning quizzes: Unassessed. Posted on Fridays

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- But make sure you check and understand it first

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- An Introduction to Statistical Learning
- Interactively exploring high-dimensional data and models in R
- Interpretable Machine Learning
- Hands-On Machine Learning with R

How to do well

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- **Ask** questions

Looking forward

Machine learning is a big, big area. This semester is like the tip of the iceberg, there's a lot more, and interesting methods and problems, than what we can cover. Take this as a challenge to get you started, and become hungry to learn more!

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- This class is an Introduction to Machine Learning
- I will also Lecture **ETC3555/5555 Statistical Machine Learning** next semester where we will have a chance to dive deeper into some topics

In todays class

- Look at different types of learning problems

We will continually refer back to the topics introduced in this class throughout the semester

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- Introduce your first machine learning model - k-nearest neighbours
- Establish a framework for good machine learning modelling

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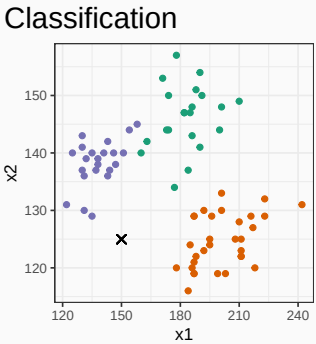
Types of problems

Framing the problem

1 **Supervised classification**: categorical $y_i \in \{1, \ldots, C\}$ is available for all $x_i \in \mathbb{R}^d$



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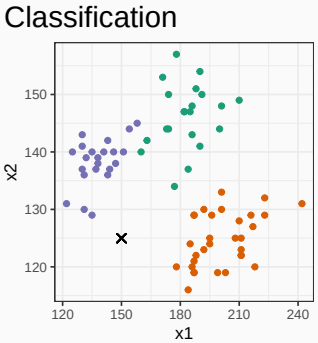

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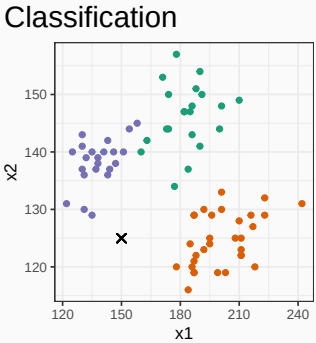

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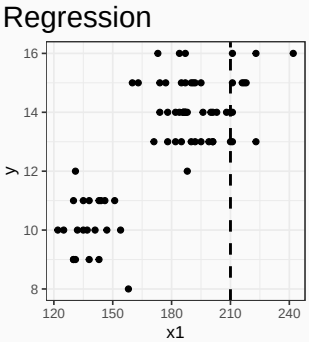

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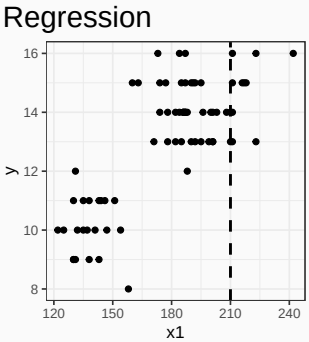

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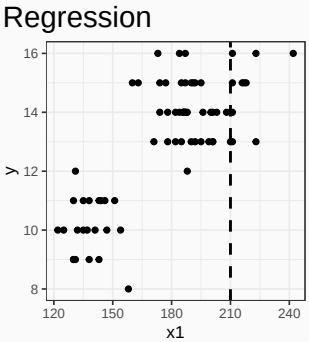

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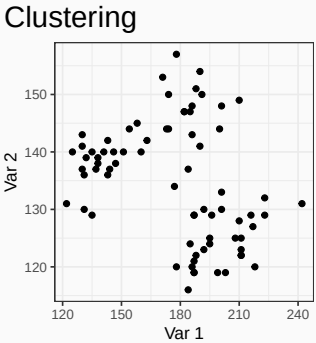

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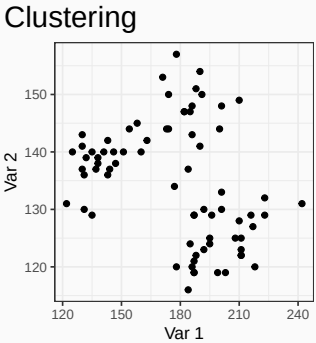

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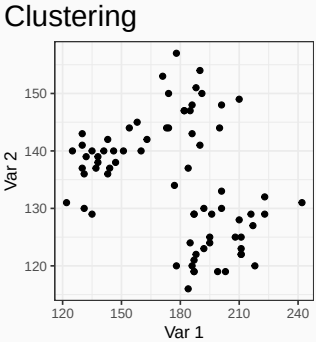
- Uncover some latent structure (e.g. clustering)



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- Every person monitored their email for a week and recorded information about each email message; for example, whether it was spam, and what day of the week and time of day the email arrived.
- We want to use this information to build a spam filter, a classifier that will catch spam with high probability but will never classify good email as spam.

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- Problem 2 was a classification problem. We wanted to predict a discrete response, i.e. whether an email was spam or not, from data
- Problem 3 was an unsupervised clustering problem. We wanted to identify different attributes of different sections of the population in order to tailor different products to them

Math and computing foundations

Predictors: math

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We will define $\mathcal{X} \subset \mathbb{R}^p$ as the set in which the predictors live.

Predictors: computing

```r
library(mvtnorm)
vc <- matrix(c(1, 0.5, 0.2,
               0.5, 1, -0.3,
               0.2, -0.3, 1),
             ncol=3, byrow=TRUE)
set.seed(449)
x <- rmvnorm(5,
             mean = c(-0.2, 0, 0.3),
             sigma = vc)
x
```

```
      [,1]  [,2]  [,3]
[1,] -0.42  1.01 -0.64
[2,] -0.83 -0.99 -0.37
[3,] -0.98  0.66 -0.13
[4,] -0.13 -0.25  0.57
[5,] -0.37  0.55  1.37
```

What's the dimension of the data?

```r
library(palmerpenguins)
p_tidy <- penguins |>
  select(species, bill_length_mm:body_mass_
  rename(bl=bill_length_mm,
         bd=bill_depth_mm,
         fl=flipper_length_mm,
         bm=body_mass_g)
p_tidy |> slice_head(n=10)
```

```
# A tibble: 10 x 5
   species    bl    bd    fl    bm
   <fct>   <dbl> <dbl> <int> <int>
 1 Adelie   39.1  18.7   181  3750
 2 Adelie   39.5  17.4   186  3800
 3 Adelie   40.3  18     195  3250
 4 Adelie   NA    NA      NA    NA
 5 Adelie   36.7  19.3   193  3450
 6 Adelie   39.3  20.6   190  3650
 7 Adelie   38.9  17.8   181  3625
```

R's pipe

A reminder for anyone who is not familiar with R's pipe

```r
library(palmerpenguins)
p_tidy1 <- penguins |>
  select(species, bill_length_mm:body_mass
  rename(bl=bill_length_mm,
         bd=bill_depth_mm,
         fl=flipper_length_mm,
         bm=body_mass_g)
p_tidy1 |> slice_head(n=10)
```

```
# A tibble: 10 x 5
   species    bl    bd    fl    bm
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 3 Adelie   40.3  18     195  3250
 4 Adelie   NA    NA      NA    NA
 5 Adelie   36.7  19.3   193  3450
```

```r
library(palmerpenguins)
p_tidy1 <- select(penguins, species, bill_
p_tidy2 <- rename(p_tidy1,
         bl=bill_length_mm,
         bd=bill_depth_mm,
         fl=flipper_length_mm,
         bm=body_mass_g)
slice_head(p_tidy2, n=10)
```

```
# A tibble: 10 x 5
   species    bl    bd    fl    bm
   <fct>   <dbl> <dbl> <int> <int>
 1 Adelie   39.1  18.7   181  3750
 2 Adelie   39.5  17.4   186  3800
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 4 Adelie   NA    NA      NA    NA
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```r
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p_tidy1 <- select(penguins, species, bill_
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         bl=bill_length_mm,
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```
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 4 Adelie   NA    NA      NA    NA
 5 Adelie   36.7  19.3   193  3450
```

Observations and variables: math

The ith observation is denoted as

$$x_i = \begin{pmatrix} x_{i1} & x_{i2} & \dots & x_{ip} \end{pmatrix}$$

The jth variable is denoted as

$$x_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix}$$

Observations and variables: computing

Observations - rows

```r
x[2,]
```

```
[1] -0.83 -0.99 -0.37
```

```r
p_tidy |> slice_sample()
```

```
# A tibble: 1 x 5
  species      bl    bd    fl    bm
  <fct>     <dbl> <dbl> <int> <int>
1 Chinstrap  51.5  18.7   187  3250
```

```r
p_tidy |> slice(2)
```

```
# A tibble: 1 x 5
  species    bl    bd    fl    bm
  <fct>   <dbl> <dbl> <int> <int>
```

Variables - columns

```r
x[,1]
```

```
[1] -0.42 -0.83 -0.98 -0.13 -0.37
```

```r
p_tidy |> pull(fl)
```

```
  [1] 181 186 195  NA 193 190 181 195 193
 [14] 191 198 185 195 197 184 194 174 180
 [27] 183 187 172 180 178 178 188 184 195
 [40] 184 182 195 186 196 185 190 182 179
 [53] 190 200 187 191 186 193 181 194 185
 [66] 192 195 188 190 198 190 190 196 197
 [79] 187 195 189 196 187 193 191 194 190
 [92] 205 185 186 187 208 190 196 178 192
[105] 193 184 199 190 181 197 198 191 193
[118] 199 189 189 187 198 176 202 186 199
[131] 190 197 193 199 187 190 191 200 185
```

Response: math

The response variable (or target variable, output, outcome measurement), when it exists, is denoted as:

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We will define $\mathcal{Y}$ as the set in which the responses live.

The whole dataset can also be written as

$$\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n = \{(y_1, x_1), (y_2, x_2), \ldots, (y_n, x_n)\} \in \{\mathcal{X} \times \mathcal{Y}\}^n$$

where x_i is a vector with p elements.

Response: computing

`species` is the response variable, and it is categorical.

```
set.seed(424)
p_tidy |> slice_sample(n=10)
```

```
# A tibble: 10 x 5
   species       bl    bd    fl    bm
   <fct>      <dbl> <dbl> <int> <int>
 1 Gentoo      47.4  14.6   212  4725
 2 Gentoo      46.2  14.5   209  4800
 3 Chinstrap   45.6  19.4   194  3525
 4 Adelie      38.8  17.6   191  3275
 5 Gentoo      50    15.3   220  5550
 6 Chinstrap   46    18.9   195  4150
 7 Adelie      38.9  18.8   190  3600
 8 Gentoo      46.5  14.5   213  4400
 9 Gentoo      46.8  14.3   215  4850
10 Gentoo      45.7  13.9   214  4400
```

A binary matrix format is sometimes useful.

```
set.seed(424)
model.matrix(~ 0 + species, data = p_tidy)
  as_tibble() |>
  slice_sample(n=10)
```

```
# A tibble: 10 x 3
   speciesAdelie speciesChinstrap speciesG
           <dbl>            <dbl>
 1             0                0
 2             0                0
 3             0                1
 4             1                0
 5             0                0
 6             0                1
 7             1                0
 8             0                0
```

Response: computing

`species` is the response variable, and it is categorical.

```
set.seed(424)
p_tidy |> slice_sample(n=10)
```

```
# A tibble: 10 x 5
   species       bl    bd    fl    bm
   <fct>      <dbl> <dbl> <int> <int>
 1 Gentoo      47.4  14.6   212  4725
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```

A binary matrix format is sometimes useful.

```
set.seed(424)
model.matrix(~ 0 + species, data = p_tidy)
  as_tibble() |>
  slice_sample(n=10)
```

```
# A tibble: 10 x 3
   speciesAdelie speciesChinstrap speciesG
           <dbl>            <dbl>
 1             0                0
 2             0                0
 3             0                1
 4             1                0
 5             0                0
 6             0                1
 7             1                0
 8             0                0
```

Linear algebra

A transposed data matrix is denoted as

$$\mathbf{X}^\top = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{n1} \\ x_{12} & x_{22} & \dots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1p} & x_{2p} & \dots & x_{np} \end{pmatrix}_{p \times n}$$

```r
x
```

```
      [,1]  [,2]  [,3]
[1,] -0.42  1.01 -0.64
[2,] -0.83 -0.99 -0.37
[3,] -0.98  0.66 -0.13
[4,] -0.13 -0.25  0.57
[5,] -0.37  0.55  1.37
```

```r
t(x)
```

```
      [,1]  [,2]  [,3]  [,4]  [,5]
[1,] -0.42 -0.83 -0.98 -0.13 -0.37
[2,]  1.01 -0.99  0.66 -0.25  0.55
[3,] -0.64 -0.37 -0.13  0.57  1.37
```

Matrix multiplication: math

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

Matrix multiplication: math

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

$$\mathbf{B}_{3\times 4} = \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{pmatrix}$$

Matrix multiplication: math

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

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then

$$\mathbf{AB}_{2\times 4} = \begin{pmatrix} \sum_{j=1}^{3} a_{1j}b_{j1} & \sum_{j=1}^{3} a_{1j}b_{j2} & \sum_{j=1}^{3} a_{1j}b_{j3} & \sum_{j=1}^{3} a_{1j}b_{j4} \\ \sum_{j=1}^{3} a_{2j}b_{j1} & \sum_{j=1}^{3} a_{2j}b_{j2} & \sum_{j=1}^{3} a_{2j}b_{j3} & \sum_{j=1}^{3} a_{2j}b_{j4} \end{pmatrix}$$

Matrix multiplication: math

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

$$\mathbf{B}_{3\times 4} = \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{pmatrix}$$

then

$$\mathbf{AB}_{2\times 4} = \begin{pmatrix} \sum_{j=1}^{3} a_{1j}b_{j1} & \sum_{j=1}^{3} a_{1j}b_{j2} & \sum_{j=1}^{3} a_{1j}b_{j3} & \sum_{j=1}^{3} a_{1j}b_{j4} \\ \sum_{j=1}^{3} a_{2j}b_{j1} & \sum_{j=1}^{3} a_{2j}b_{j2} & \sum_{j=1}^{3} a_{2j}b_{j3} & \sum_{j=1}^{3} a_{2j}b_{j4} \end{pmatrix}$$

Pour the rows into the columns.

Matrix multiplication: math

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

$$\mathbf{B}_{3\times 4} = \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{pmatrix}$$

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Pour the rows into the columns.

Note: You can't do **BA**!

Matrix multiplication: computing

```r
x
```

```
      [,1]  [,2]  [,3]
[1,] -0.42  1.01 -0.64
[2,] -0.83 -0.99 -0.37
[3,] -0.98  0.66 -0.13
[4,] -0.13 -0.25  0.57
[5,] -0.37  0.55  1.37
```

```r
B <- matrix(c(1/sqrt(2), 1/sqrt(2), 0,
              0, 0, 1), ncol=2, byrow=FALSE)
B
```

```
     [,1] [,2]
[1,] 0.71    0
[2,] 0.71    0
[3,] 0.00    1
```

Try this:

```r
t(x) %*% B
```

It produces an error because it can't be done

```
Error in t(x) %*% B : non-conformable argu
```

Notice: `%*%` uses a `*` so it is NOT the tidyverse pipe.

Identity matrix

$$I = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & 0 & & 1 \end{pmatrix}_{p \times p}$$

```r
diag(1, 8, 8)
```

```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,
[1,]    1    0    0    0    0    0    0
[2,]    0    1    0    0    0    0    0
[3,]    0    0    1    0    0    0    0
[4,]    0    0    0    1    0    0    0
[5,]    0    0    0    0    1    0    0
[6,]    0    0    0    0    0    1    0
[7,]    0    0    0    0    0    0    1
[8,]    0    0    0    0    0    0    0
```

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[2,]    0    1    0    0    0    0    0
[3,]    0    0    1    0    0    0    0
[4,]    0    0    0    1    0    0    0
[5,]    0    0    0    0    1    0    0
[6,]    0    0    0    0    0    1    0
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```

This is the matrix equivalent of 1

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```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,
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[2,]    0    1    0    0    0    0    0
[3,]    0    0    1    0    0    0    0
[4,]    0    0    0    1    0    0    0
[5,]    0    0    0    0    1    0    0
[6,]    0    0    0    0    0    1    0
[7,]    0    0    0    0    0    0    1
[8,]    0    0    0    0    0    0    0
```

This is the matrix equivalent of 1

$AI = A$ and $IA = A$ (as long as A was conformable with these matrix multiplications)

Inverting a matrix: math

Suppose that **A** is square

$$\mathbf{A}_{2\times 2} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

then the inverse is (if $ad - bc \neq 0$)

$$\mathbf{A}^{-1}_{2\times 2} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

and $\mathbf{A}\mathbf{A}^{-1} = I$ where

$$\mathbf{I}_{2\times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

If $AB = I$, then $B = A^{-1}$.

```
vc
```

```
     [,1] [,2] [,3]
[1,]  1.0  0.5  0.2
[2,]  0.5  1.0 -0.3
[3,]  0.2 -0.3  1.0
```

```
vc_i <- solve(vc)
vc_i
```

```
      [,1]  [,2]  [,3]
[1,]  1.62 -1.00 -0.63
[2,] -1.00  1.71  0.71
[3,] -0.63  0.71  1.34
```

```
vc %*% vc_i
```

k-nearest neighbours

k-nearest neighbours

k-nearest neighbours (kNN) is possibly the simplest non-linear non-parametric supervised learning model

Idea: For any new observation x' with unknown response (class or value), predict its response based on the average responses $\{y_i\}_{i=1}^n$ of K training set observations,
whose features $\{x_i\}_{i=1}^n$ are closest to x'

More like a recipe/algorithm for prediction than a model

All you need to specify is

- A distance $d(x, x_k)$

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More like a recipe/algorithm for prediction than a model

All you need to specify is

- A distance $d(x, x_k)$
- A number of neighbours K

Recipe for prediction

- Specify a distance metric $d(\cdot, \cdot)$ (usually defaults to $d(x_1, x_2) = \sqrt{\sum_{j=1}^{p}(x_{1j} - x_{2j})^2}$) and a number of neighbours K

Recipe for prediction

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  - for $i = 1, \ldots, n$ evaluate $d(x', x_i)$

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    - In regression predict $\frac{1}{K}\sum_{k=1}^{K} y_{n_k}$

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- When predicting new observation x'
  - for $i = 1, \ldots, n$ evaluate $d(x', x_i)$
  - select the K observations with the smallest $d(x', x_i)$
  - Predict the response as the average of the y_i of the K nearest neighbours
    - In regression predict $\frac{1}{K}\sum_{k=1}^{K} y_{n_k}$
    - In classification predict $P(y = C) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{I}(y_{n_k} = C)$

Predicting Probabilities

In classification, the majority of machine learning models will not be a functions that directly produces prediction - $\widehat{y} : \mathcal{X} \mapsto \{1, \ldots, C\}$

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Let these probabilities be $\widehat{p} : \mathcal{X} \mapsto (0, 1)^C$ with $\sum_{c=1}^{C} \widehat{p}(x_i)_c = 1$ where $\widehat{p}(x_i)_c = \mathbb{P}(y = c \mid x_i)$

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Let these probabilities be $\widehat{p} : \mathcal{X} \mapsto (0, 1)^C$ with $\sum_{c=1}^{C} \widehat{p}(x_i)_c = 1$ where $\widehat{p}(x_i)_c = \mathbb{P}(y = c \mid x_i)$

We can always turn these probabilities into class prediction i.e. $\widehat{y}(x_i) = \arg\max \widehat{p}(x_i)_c$

But the probabilities themselves contain much more information than just a class labels

Predicting Probabilities

In classification, the majority of machine learning models will not be a functions that directly produces prediction - $\widehat{y} : \mathcal{X} \mapsto \{1, \ldots, C\}$

We will see in this class that it is much better to estimate probabilities of being in each class.

Let these probabilities be $\widehat{p} : \mathcal{X} \mapsto (0, 1)^C$ with $\sum_{c=1}^{C} \widehat{p}(x_i)_c = 1$ where $\widehat{p}(x_i)_c = \mathbb{P}(y = c \mid x_i)$

We can always turn these probabilities into class prediction i.e. $\widehat{y}(x_i) = \arg\max \widehat{p}(x_i)_c$

But the probabilities themselves contain much more information than just a class labels

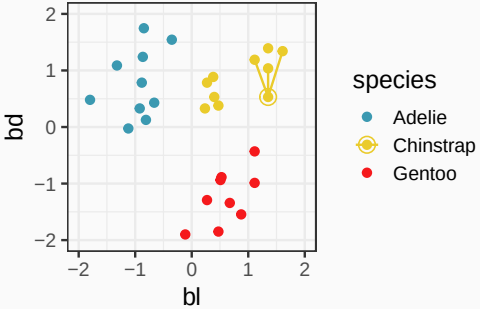
And it is easier to train a model to learn them

k-nearest neighbours

Predict y using the $k-$ nearest neighbours from observation of interest.

$\hat{P}(y = \text{Chinstrap}) = 1$
$\hat{P}(y = \text{Gentoo}) = 0$
$\hat{P}(y = \text{Adelie}) = 0$

K=3

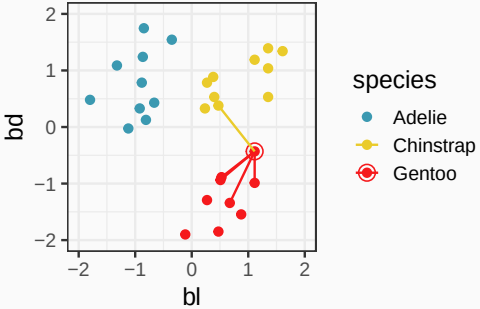


k-nearest neighbours

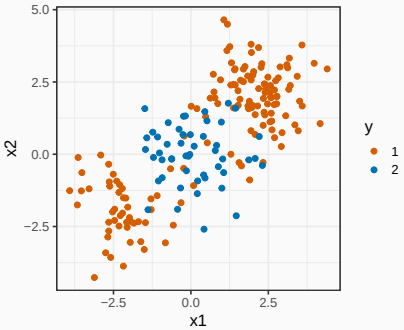
Predict y using the $k-$ nearest neighbours from observation of interest.

$\hat{P}(y = \text{Chinstrap}) = \frac{1}{5}$
$\hat{P}(y = \text{Gentoo}) = \frac{4}{5}$
$\hat{P}(y = \text{Adelie}) = 0$

K=5



Implementation




```r
set.seed(3)
sim_data2 |> slice_sample(n = 5)
```

```
# A tibble: 5 x 3
       x1      x2 y
   <dbl>   <dbl> <fct>
1  2.33  -0.285  1
2 -0.159 -0.0565 2
3 -2.06  -2.53   1
4  1.59   1.90   1
5  0.412 -0.715  2
```

Implementation

Throughout this course we are going to use the `tidymodels` syntax for fitting our machine learning models https://parsnip.tidymodels.org/reference/nearest_neighbor.html

```
K <- 5
```

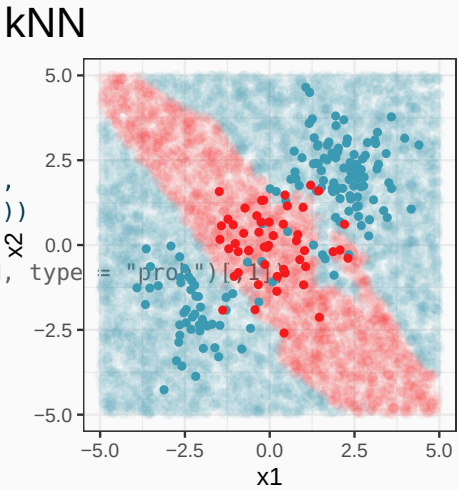
```
knn_spec <- nearest_neighbor(neighbors = K, weight_func = "recta
  set_mode("classification") |>
  set_engine("kknn") #specify how to learn the parameters
```

```
knn_fit <- knn_spec |> ## apply the data
  fit(y ~ ., data = sim_data2)
```

Results

```
sim_data2_grid <- tibble(
  x1 = runif(10000, -5, 5),
  x2 = runif(10000, -5, 5))


sim_data2_grid <- sim_data2_grid |>
  mutate(cl_knn = factor(predict(knn_fit$fit,
                                 sim_data2_grid)))
  #mutate(prob_knn = predict(knn_fit$fit,
  #                          sim_data2_grid, type = "prob")[,1])
```



Pros and Cons

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- Very simple,

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Cons

- Distance gives all features the same weight,
- There is no automatic way to give important features increased weight or learn feature importance
- Fails in high dimensions because data is too sparse - in these cases the model needs some structure (assumptions)

Evaluating your classification model

Training vs test splits

When data are reused for multiple tasks, instead of carefully *spent* from the finite data budget, certain risks increase, such as the risk of accentuating bias or compounding effects from methodological errors. *Julia Silge*

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Therefore we split the data and conduct different tasks on different subsets of data

- Training set $\approx$ 80%: Used to fit the model. This might also be broken into a validation set(s) for hyperparameter tuning and/or frequent assessment of fit.

Training vs test splits

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Therefore we split the data and conduct different tasks on different subsets of data

- Training set $\approx$ 80%: Used to fit the model. This might also be broken into a validation set(s) for hyperparameter tuning and/or frequent assessment of fit.
- Test set $\approx$ 20%: Purely used to assess final models performance on future data.

Training vs test splits

DO NOT only evaluate your model on the training set.

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The model was estimated on that training set so it is designed to predict that set well. (kNN with $K = 1$ predicts the training data perfectly)

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The model was estimated on that training set so it is designed to predict that set well. (kNN with $K = 1$ predicts the training data perfectly)

Therefore this will give you a **BIASED** estimate of its performance.

In particular a more complex model will always *APPEAR* to predict the training set better

Training vs test splits

Figure 1.5 Graphs of the root-mean-square error, defined by (1.3), evaluated on the training set and on an independent test set for various values of M.

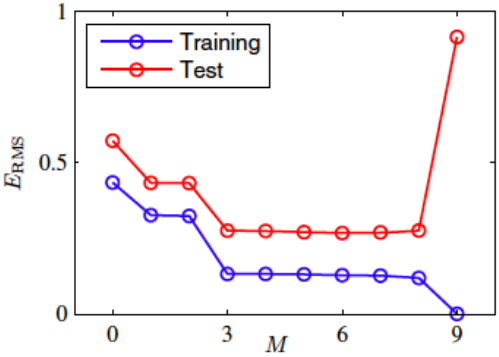

Figure 1: From 'Pattern Recognition and Machine Learning' by Christopher Bishop

Training vs test splits

```
d_bal <- tibble(y=c(rep("A", 6), rep("B",
                x=c(runif(12)))
d_bal$y
```

```
[1] "A" "A" "A" "A" "A" "A" "B" "B" "B" "B"
```

```
d_unb <- tibble(y=c(rep("A", 2), rep("B",
                x=c(runif(12)))
d_unb$y
```

```
[1] "B" "A" "B" "A" "B" "B" "B" "B" "B" "B" "B" "
```

```
set.seed(130)
d_bal_split <- initial_split(d_bal, prop =
training(d_bal_split)$y
```

```
[1] "A" "A" "B" "A" "B" "A" "B" "A"
```

```
set.seed(132)
d_unb_split <- initial_split(d_unb, prop =
training(d_unb_split)$y
```

```
[1] "B" "B" "A" "B" "B" "A" "B" "B"
```

```
testing(d_bal_split)$y
```

```
[1] "A" "B" "B" "B"
```

```
testing(d_unb_split)$y
```

```
[1] "B" "B" "B" "B"
```

Setting the seed ensures that if we

Training vs test splits

```
d_unb <- tibble(y=c(rep("A", 2), rep("B", 10)),
                x=c(runif(12)))
d_unb$y
```

```
 [1] "A" "A" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
```

```
set.seed(132)
d_unb_split <- initial_split(d_unb, prop = 0.70)
training(d_unb_split)$y
```

```
[1] "B" "B" "A" "B" "B" "A" "B" "B"
```

```
testing(d_unb_split)$y
```

```
[1] "B" "B" "B" "B"
```

Always stratify splitting by sub-groups, especially response variable classes.

```
d_unb_strata <- initial_split(d_unb, prop =
training(d_unb_strata)$y
```

```
[1] "A" "B" "B" "B" "B" "B" "B" "B"
```

```
testing(d_unb_strata)$y
```

```
[1] "A" "B" "B" "B"
```

This ensure that your training set adequately resembles your testing set.

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While it is easier, and more informative, to model probabilities, eventually a decision/prediction must be made.

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However if the probability that a relative has cancer is 0.45, do you want the doctor to predict they don't have cancer??

The value of α is problem specific and depends on the consequences of missclassification.

It is not our job as data scientists to decide on α, we model the probabilities and let a domain expert turn this into a decision.

Measuring accuracy for categorical response

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Learn $\widehat{y}$ using $\{(y_i, \mathbf{x}_i)\}_{i=1}^{n_{train}}$

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Learn $\widehat{y}$ using $\{(y_i, \mathbf{x}_i)\}_{i=1}^{n_{train}}$

And evaluate the error rate using $\{(y_i, \mathbf{x}_i)\}_{i=n_{train}}^{n}$

Training error will usually be smaller than test error. When it is much smaller, it indicates that the model is too well fitted to the training data to be accurate on future data (over-fitted).

Confusion (misclassification) matrix

You can visualise this classification performance using a confusion matrix

predicted

`1`

`0`

true

`1`

a

b

`0`

c

d

Consider `1`=positive (P), `0`=negative (N).

- True positive (TP): a

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- False positive (FP): `c` (Type I error)

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- Sensitivity, recall, hit rate, or true positive rate (TPR): TP/P = `a`/(`a`+`b`)

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- Sensitivity, recall, hit rate, or true positive rate (TPR): TP/P = `a/(a+b)`
- Specificity, selectivity or true negative rate (TNR): TN/N = `d/(c+d)`

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- Prevalence: P/(P+N) = (a+b)/(a+b+c+d)
- Accuracy: (TP+TN)/(P+N) = (a+d)/(a+b+c+d)
- Balanced accuracy: (TPR + TNR)/2 (or average class errors)

Confusion (misclassification) matrix: computing

Two classes

```
# Write out the confusion matrix in standa
#| label: oconfusion-matrix-tidy
cm <- a2 |> dplyr::count(y, pred) |>
  group_by(y) |>
  mutate(cl_acc = n[pred==y]/sum(n))
cm |>
  pivot_wider(names_from = pred,
              values_from = n) |>
  select(y, bilby, quokka, cl_acc)
```

```
# A tibble: 2 x 4
# Groups:   y [2]
  y      bilby quokka cl_acc
  <fct>  <int>  <int>  <dbl>
1 bilby      9      3  0.75
2 quokka     5     10  0.667
```

More than two classes

```
# Write out the confusion matrix in standa
cm3 <- a3 |> dplyr::count(y, pred) |>
  group_by(y) |>
  mutate(cl_err = n[pred==y]/sum(n))
cm3 |>
  pivot_wider(names_from = pred,
              values_from = n, values_fill
  select(y, bilby, quokka, numbat, cl_err)
```

```
# A tibble: 3 x 5
# Groups:   y [3]
  y      bilby quokka numbat cl_err
  <fct>  <int>  <int>  <int>  <dbl>
1 bilby      9      3      0  0.75
2 numbat     0      2      6  0.75
3 quokka     5     10      0  0.667
```

Receiver Operator Curves (ROC)

However $\hat{y}$ required a value of α, and we said this was problem dependent and not our job to set

Receiver Operator Curves (ROC)

However $\hat{y}$ required a value of α, and we said this was problem dependent and not our job to set

Therefore we need to be able to evaluate our models for any value of α

Evaluate TPR and FPR for different α

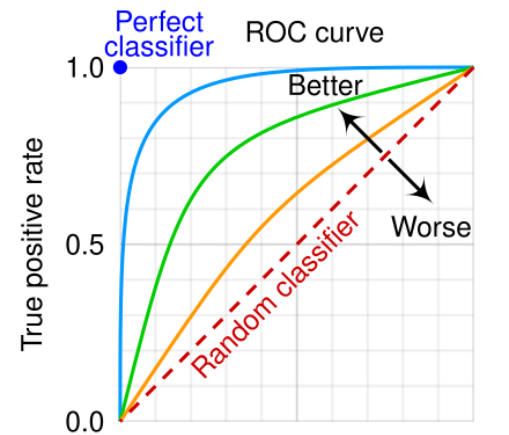
If α = 1, $\hat{p}(x_i)$ can never be greater than 1

TPR = 0, FPR = 0

If α = 0, $\hat{p}(x_i)$ always greater than 0

TPR = 1, FPR = 1

For a good classifier, as we increase α the TPR should go up faster than the FPR



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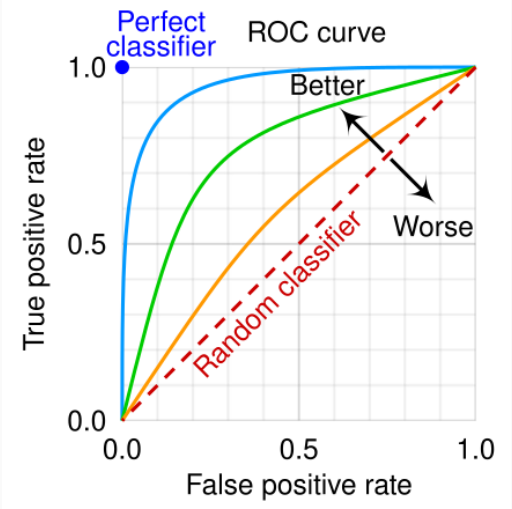
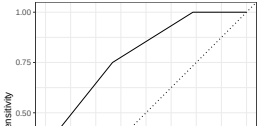



```
a2 |> slice_head(n=3)
```

```
# A tibble: 3 x 4
  y     pred  bilby quokka
  <fct> <fct> <dbl>  <dbl>
1 bilby bilby   0.9    0.1
2 bilby bilby   0.8    0.2
3 bilby bilby   0.9    0.1
```

```
roc_curve(a2, y, bilby) |>
  autoplot()
```



Conceptual framework

Accuracy vs interpretability

Why are we learning from data?

Predictive accuracy

The primary purpose is to be able to predict $\hat{y}$ for new data x'. And we'd like to do that well! That is, accurately.

	HIGH ACCURACY	MEDIUM ACCURACY	LOW ACCURACY
HIGH PRECISION	BARACK OBAMA WAS PRESIDENT FOR 70,128 HOURS	BARACK OBAMA WEIGHS AS MUCH AS 17,082 CATS	BARACK OBAMA IS 70,128 FEET TALL
MEDIUM PRECISION	MOST CATS HAVE 4 LEGS	BARACK OBAMA IS 6'1"	BARACK OBAMA HAS 4 LEGS
		BARACK	BARACK

Interpretability

Almost equally important is that we want to understand the relationship between **x** and y.

Simpler models are easier to understand. The simpler model that is (almost) as accurate is the one we choose, always.

The big picture

1 Know your data

The big picture

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   - Categorical response or no response

The big picture

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   - Types of predictors: quantitative, categorical

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   - Are you confident that your bias is low and variance is low?

Next: Visualisation